Do Now · retrieval

Lesson 097 · Little Man Computer: Branches and Loops

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does BRA do in LMC?

Q2

When does BRZ branch?

Q3

What does LDA COUNT do?

Q4

Stack Push2,Push5,Pop,Push8: final bottom-to-top?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — BRA always changes which instruction is fetched next; BRZ branches only when ACC is zero.

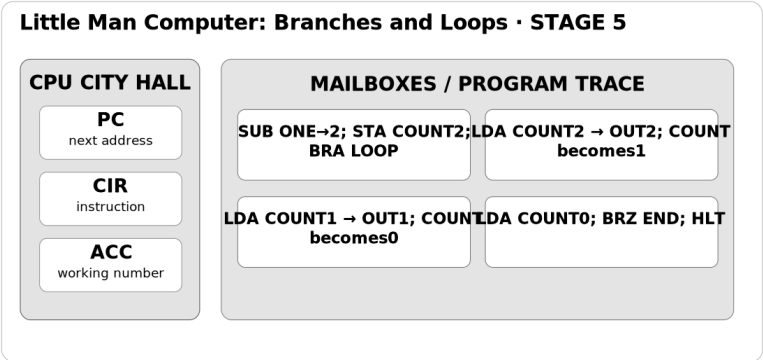
Core idea 2

Apply — The loop's memory variable COUNT changes after SUB ONE and STA COUNT. Reload it with LDA COUNT at each iteration.

Core idea 3

Evaluate — Trace ACC, COUNT, branch decision and outputs. Stopping before OUT when COUNT=0 prevents printing zero.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

BRA — Always branch to label.

BRZ — Branch if ACC zero.

label — Symbolic instruction/memory address.

loop — Repeated control flow.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

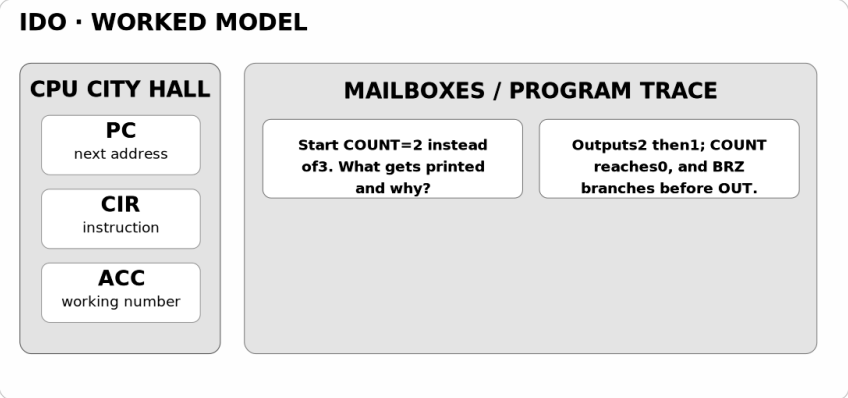
Problem

Start COUNT=2 instead of3. What gets printed and why?

Follow the worked method

1. Follow the reasoning — Outputs2 then1; COUNT reaches0, and BRZ branches before OUT.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Follow the reasoning
Outputs4 then2; stored COUNT becomes0, and BRZ exits before the next OUT.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace COUNT DAT3 loop outputs and COUNT after each SUB. [6]

Question 2

Change COUNT DAT3 to COUNT DAT2. State outputs and explain BRZ. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A pupil replaces BRZ END with BRA END at LOOP. Explain effect and repair. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

BRZ END always jumps to END regardless of ACC.

Use key vocabulary from memory

1. Explain BRA in your own words.

2. Explain BRZ in your own words.

3. Explain label in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.